



Communication Systems Computer Assignment 2

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1 Amplitude Modulation

1.1 1. Signal Generation and Bandwidth Analysis

The message signal $m(t)$ is defined as a summation of decaying and growing exponentials, windowed between $t = 4$ and $t = 8$ seconds:

$$m(t) = [e^{-6t} + e^{6t}] \cdot (u(t - 4) - u(t - 8)) \quad (1)$$

This creates a symmetric pulse-like shape. We generated this signal in MATLAB using a sampling frequency of $F_s = 20,000$ Hz. This rate satisfies the condition $F_s \gg 2(f_c + B)$, ensuring a safe margin against aliasing and providing high resolution for simulation.

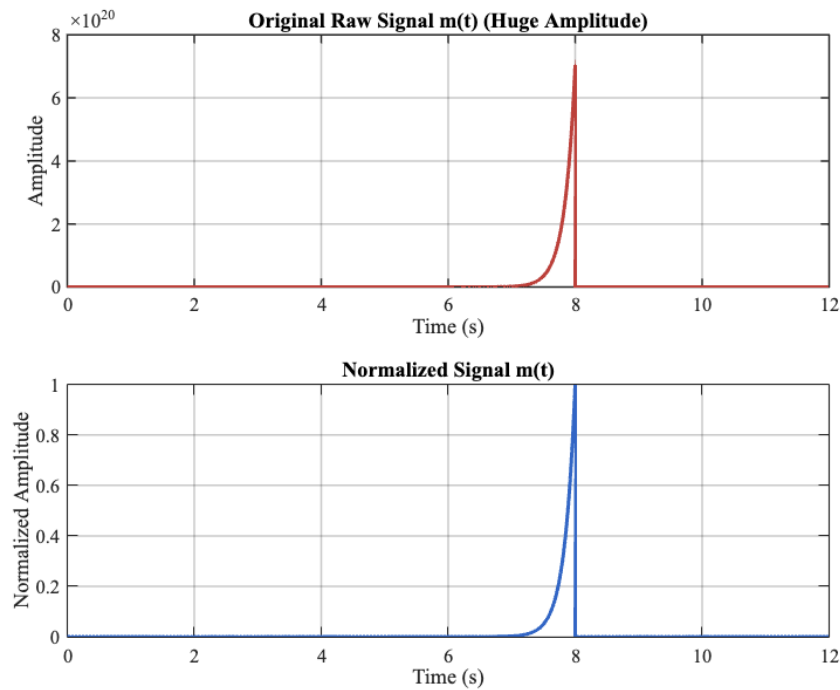
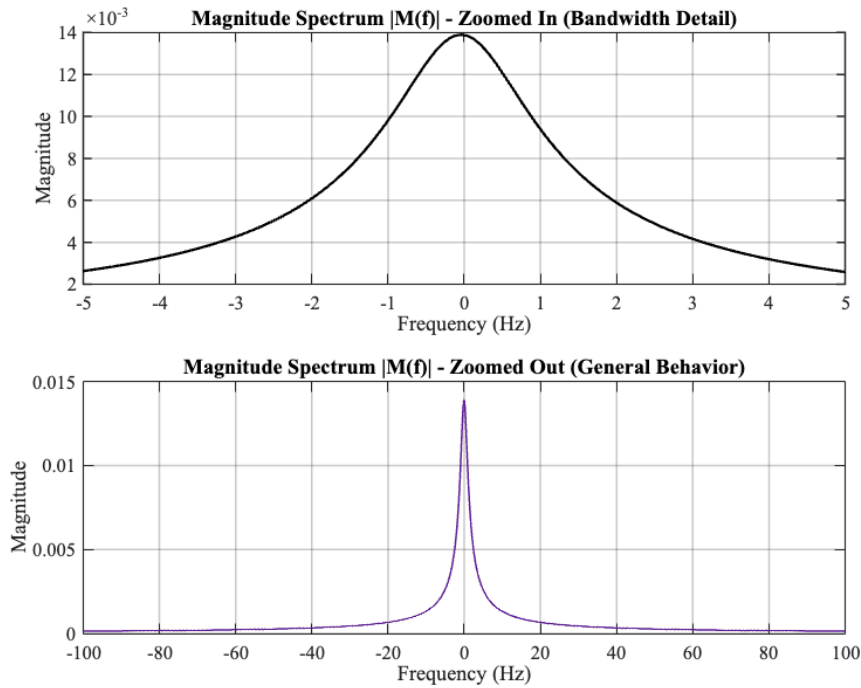


Figure 1: Time domain representation of the raw and normalized message signal $m(t)$

Figure 2: Magnitude Spectrum $|M(f)|$

Bandwidth Calculation: To determine the effective bandwidth, we utilized the 99% energy criterion. We integrated the power spectral density of the signal starting from DC ($f = 0$) and found the frequency at which the cumulative energy reached 99% of the total energy.

- **Calculated Effective Bandwidth (B_{eff}):** ≈ 60.75 Hz.
- **Nyquist Rate:** Since the Nyquist rate is defined as $2 \times BW$, the minimum sampling rate required is approximately 121.5 Hz.

1.2 2. AM Modulation

We implemented Amplitude Modulation (AM) with a carrier frequency $f_c = 20$ kHz and a modulation index $\mu = 0.85$. The carrier amplitude was set to $A_c = 1$. The expression for the AM signal is:

$$x_{AM}(t) = A_c[1 + \mu m_n(t)] \cos(2\pi f_c t) \quad (2)$$

For this part, the sampling frequency was increased to $F_s = 200,000$ Hz to accurately represent the 20 kHz carrier (10 samples per carrier period).

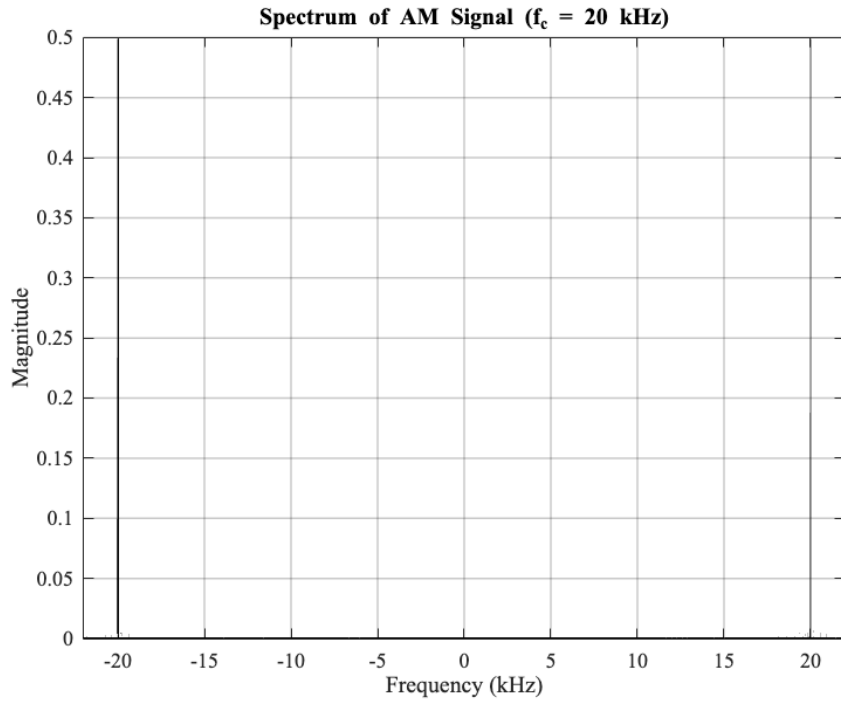


Figure 3: Spectrum of the AM modulated signal centered at $f_c = 20$ kHz.

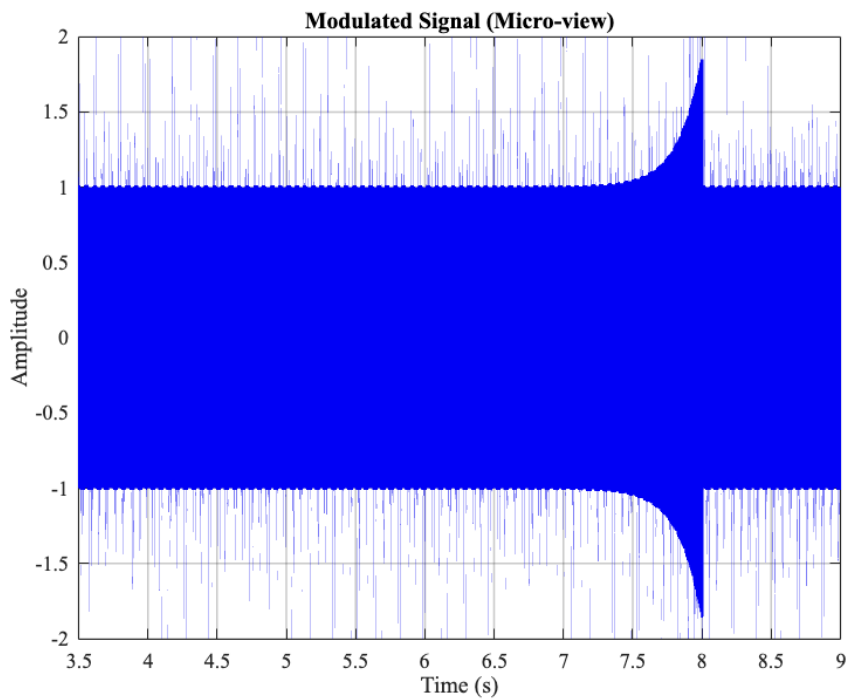


Figure 4: Time Domain view of the Modulated Signal (Zoomed).

The spectrum shows the carrier component at 20 kHz and the two sidebands (USB and LSB) containing the message information.

1.3 3. Envelope Detector Analysis

We simulated an envelope detector using a resistor-capacitor (RC) circuit model. The performance heavily relies on the time constant $\tau = RC$:

Time Constant Selection: We chose $\tau = 0.002$ s. If τ is too small, the capacitor discharges too quickly, causing excessive "ripple" at the carrier frequency. If τ is too large, the capacitor cannot discharge fast enough to follow the negative slopes of the message envelope, leading to "diagonal clipping" or failure to track the signal.

Diode Saturation Effect: In a real circuit, if the input voltage exceeds the linear range of the diode, saturation occurs. This leads to the "clipping" of signal peaks and compresses the amplitude range. This non-linearity destroys the proportional relationship between the envelope and the message, introducing significant distortion in the recovered signal.

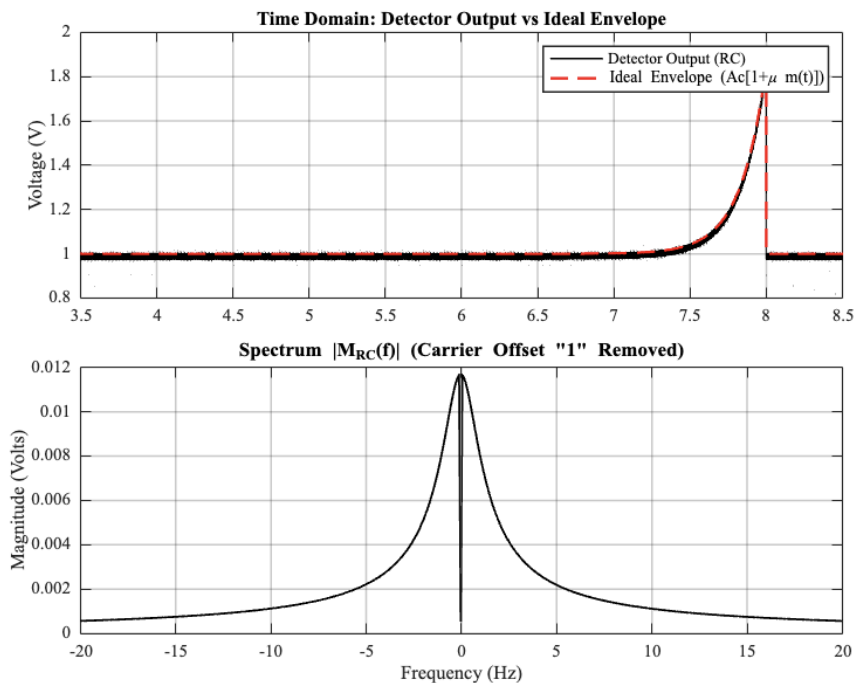


Figure 5: Envelope detector output showing the ripple effect and the comparison with the ideal envelope.

1.4 4. Rectifier and Low-Pass Filter Method

An alternative non-coherent demodulation method involves rectifying the signal and then low-pass filtering it.

$$y(t) = \text{LPF}\{|x_{AM}(t)|\} \quad (3)$$

The rectification acts as a mixing process that shifts the spectrum to baseband. For successful recovery, the filter cutoff frequency f_{cut} must satisfy:

$$B_{msg} \ll f_{cut} \ll f_c \quad (4)$$

This ensures the message is passed while the high-frequency carrier harmonics (at $2f_c, 4f_c, \dots$) are rejected. **Note on Amplitude:** The LPF often attenuates the signal energy, and

the rectification process introduces a DC offset ($\frac{2}{\pi}A_c$) and scaling factor. We corrected for these to align the recovered signal with the original.

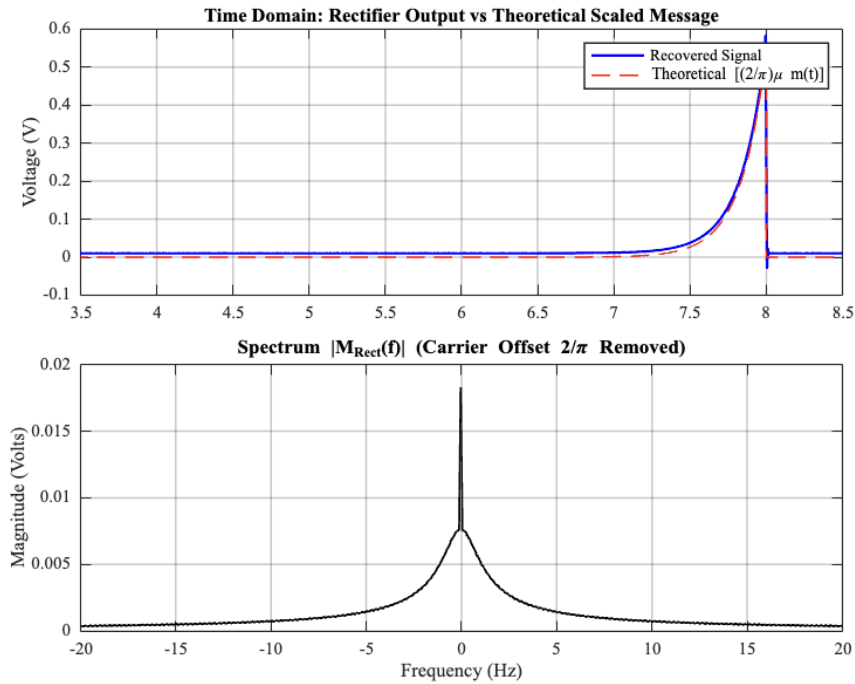


Figure 6: Rectifier output after filtering. Note the removal of high-frequency harmonics.

1.5 5. Coherent Detection

Coherent detection assumes perfect phase synchronization with the transmitter. The received signal is multiplied by a local carrier $2 \cos(2\pi f_c t)$:

$$y(t) = x_{AM}(t) \cdot 2 \cos(2\pi f_c t) \quad (5)$$

$$= A_c[1 + \mu m(t)](1 + \cos(4\pi f_c t)) \quad (6)$$

After low-pass filtering, the exact message is recovered. This method is linear and superior to envelope detection as it does not suffer from diagonal clipping or threshold effects, provided the local oscillator is perfectly synchronized (Costas Loop).

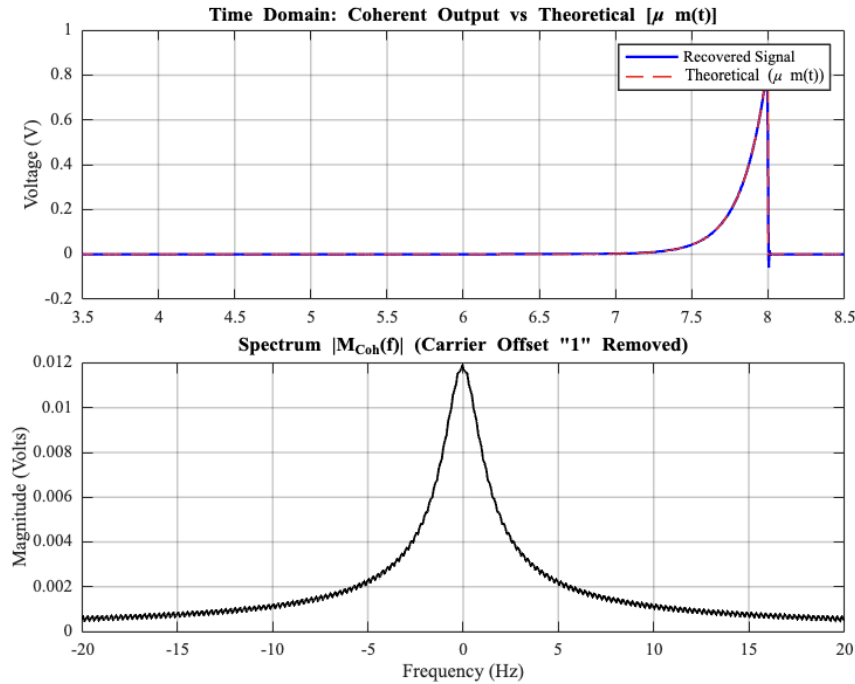


Figure 7: Coherent detection results showing near-perfect reconstruction of the original pulse.

1.6 6. Comparison of Demodulation Methods

We compared the outputs of the Envelope Detector, Rectifier, Coherent Detector, and MATLAB's 'amdemod'.

- **'amdemod' (Best Performance):** This function produced the lowest error (NMSE). It uses an optimized digital implementation that essentially performs ideal envelope detection (via Hilbert transform) or coherent detection, avoiding the hardware non-idealities like RC ripple or diode drops.
- **Coherent Detection:** Theoretically ideal, but sensitive to phase mismatch. In simulation, it performs very well.
- **Envelope/Rectifier:** These are simple and cheap to implement in hardware but suffer from inherent distortions (ripple, sensitivity to τ , non-linearity).

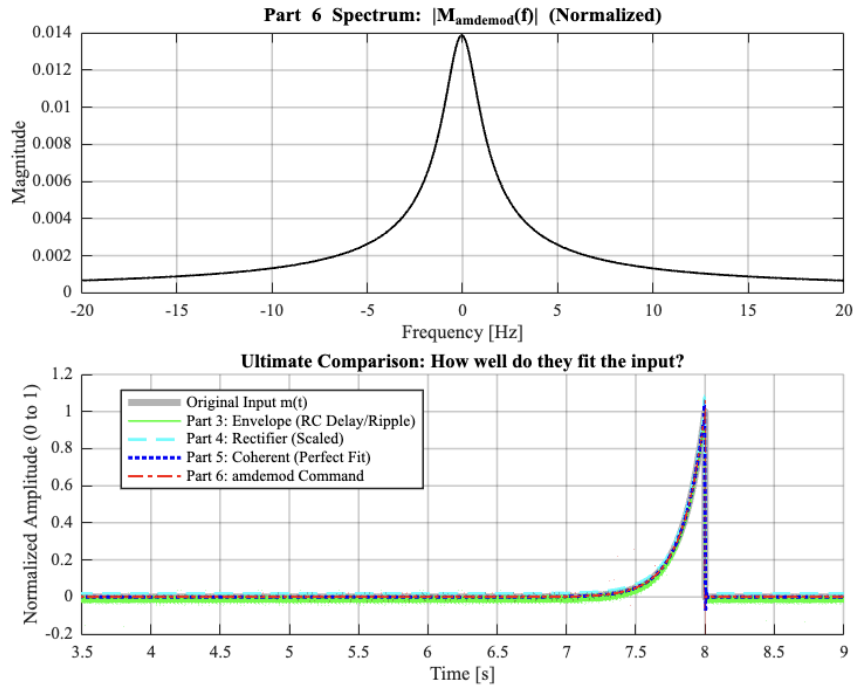


Figure 8: Overlay comparison of all demodulation techniques.

1.7 7. DSB-SC Modulation

We implemented Double Sideband Suppressed Carrier (DSB-SC). Unlike AM, the carrier is suppressed, meaning all power is dedicated to the sidebands. This improves power efficiency but necessitates coherent demodulation, as the envelope does not follow the message shape (due to phase reversals at zero crossings).

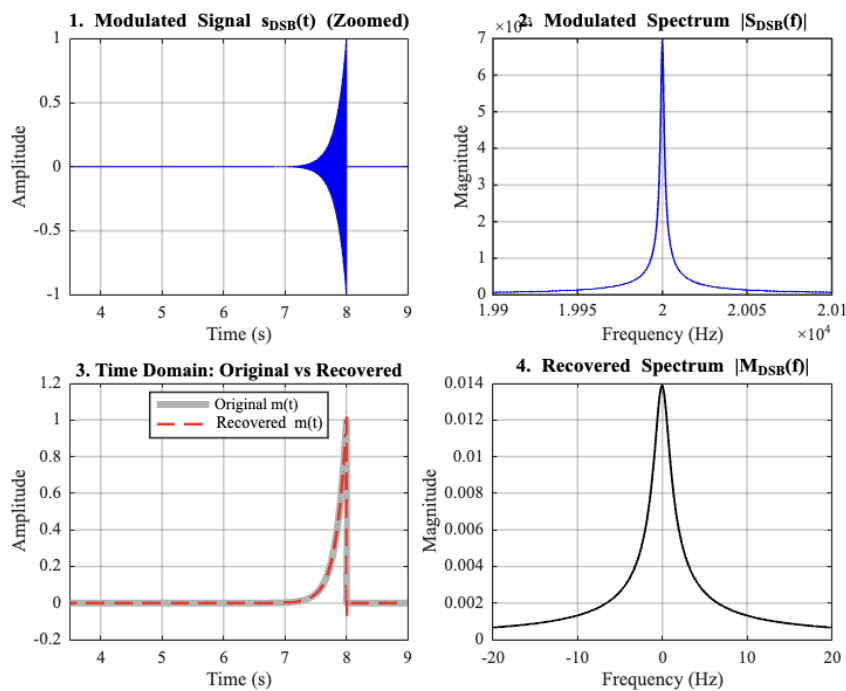


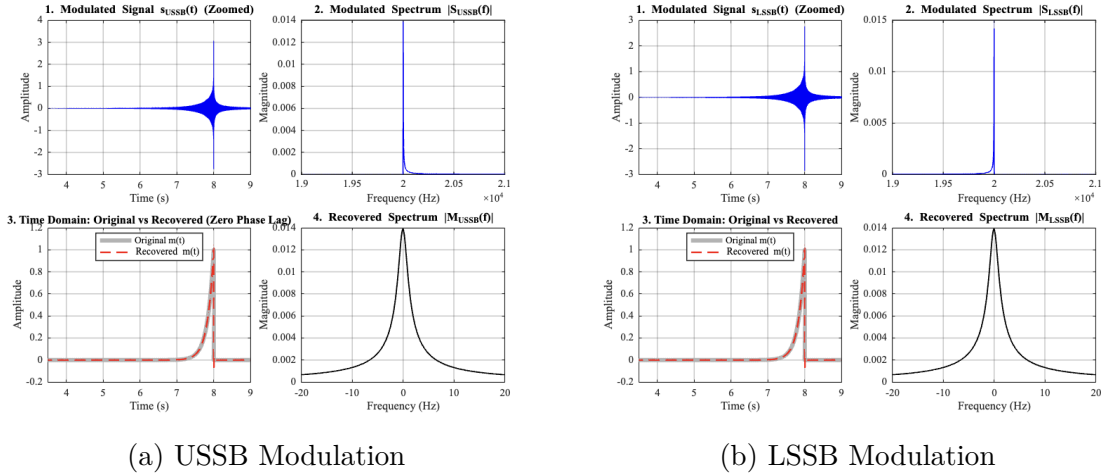
Figure 9: DSB-SC signal and its demodulation.

1.8 8 & 9. Single Sideband (SSB) Modulation

We generated USSB and LSSB signals. SSB is the most spectrally efficient method, using only half the bandwidth of AM or DSB-SC.

- **USSB:** Retains frequencies $f > f_c$.
- **LSSB:** Retains frequencies $f < f_c$.

We used coherent demodulation with zero-phase filtering to ensure a perfect time-domain match.



(a) USSB Modulation

(b) LSSB Modulation

Figure 10: SSB Modulation results showing bandwidth conservation.

1.9 10. Performance Comparison (NMSE)

We calculated the Normalized Mean Square Error (NMSE) for all modulation schemes.

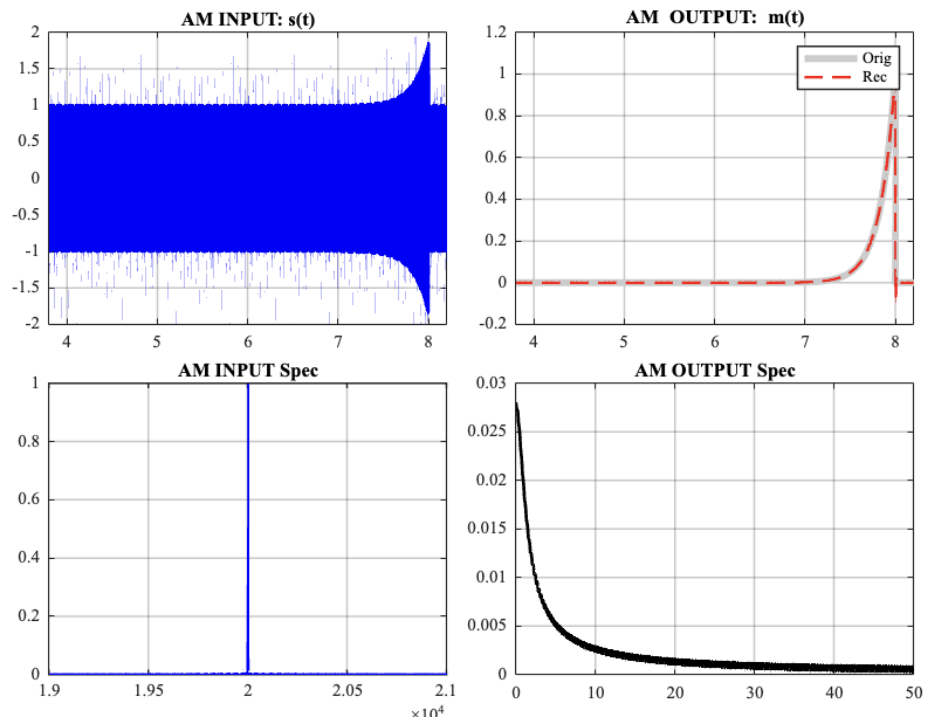


Figure 11: Detailed AM Analysis: Input $s(t)$, Output $m(t)$, Input Spec, Output Spec.

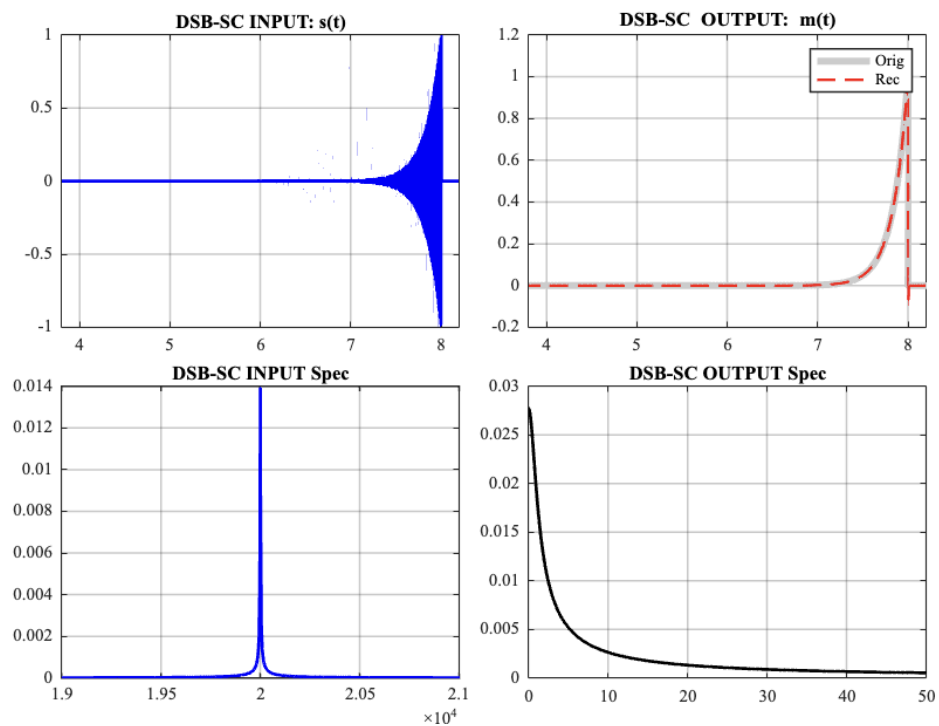


Figure 12: Detailed DSB-SC Analysis.

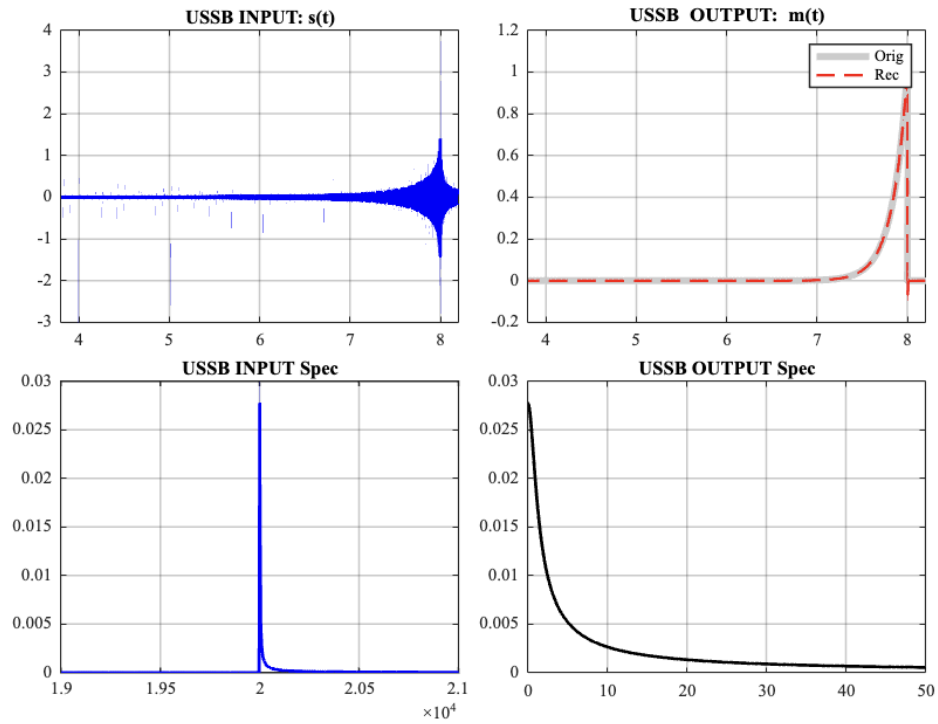


Figure 13: Detailed USSB Analysis.

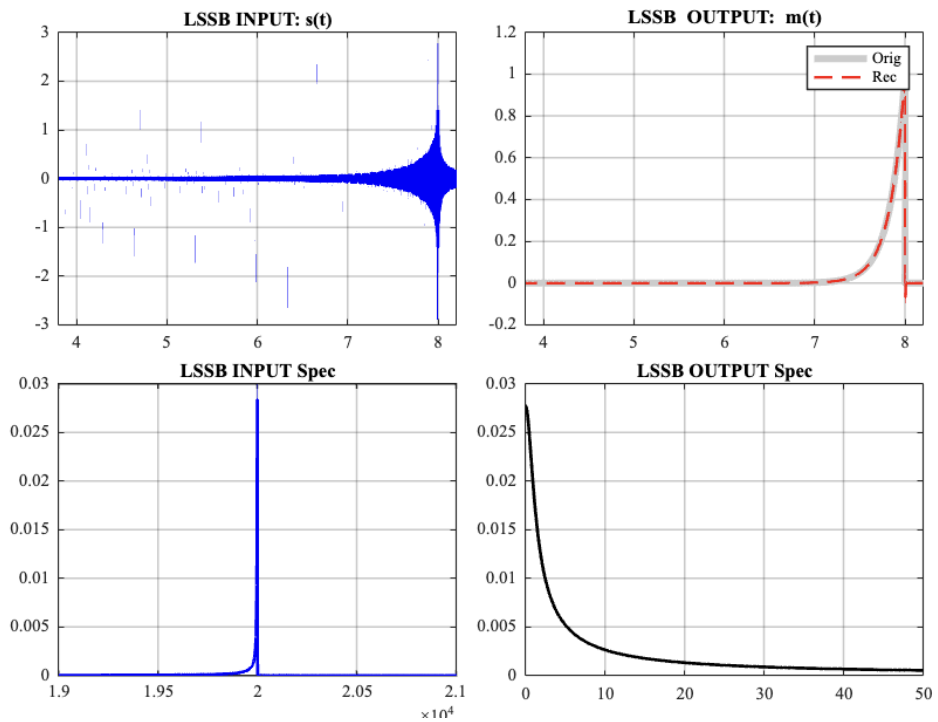


Figure 14: Detailed LSSB Analysis.

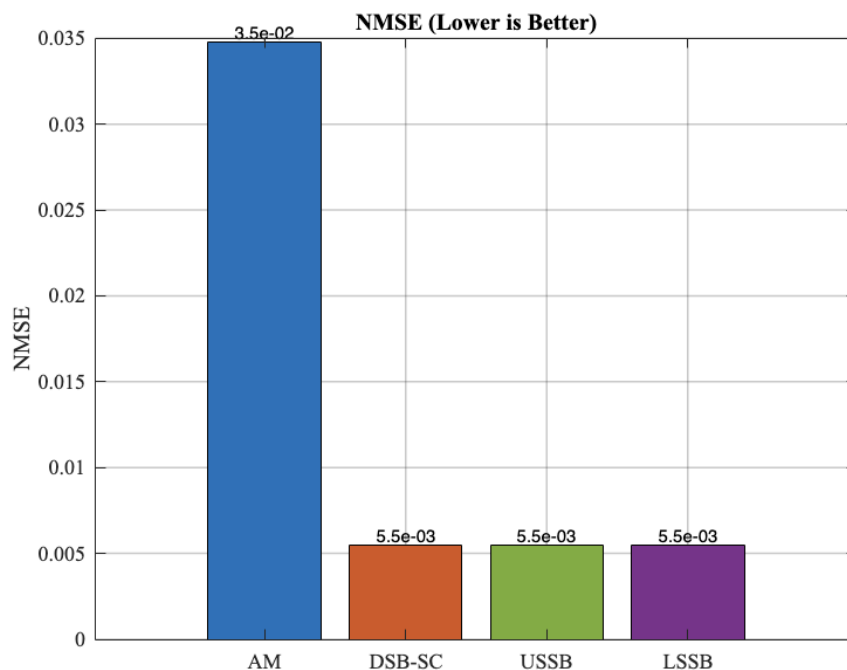


Figure 15: NMSE Bar Chart Comparison for all methods.

Method	NMSE
AM (Envelope/Rectifier)	$\approx 3.48 \times 10^{-2}$
DSB-SC	$\approx 5.46 \times 10^{-3}$
USSB	$\approx 5.46 \times 10^{-3}$
LSSB	$\approx 5.46 \times 10^{-3}$

Table 1: NMSE Comparison.

Analysis: SSB and DSB-SC achieve significantly lower errors than AM.

- **SSB superiority:** By transmitting only one sideband, SSB concentrates the signal energy more effectively. Combined with the absence of a carrier (which consumes power but carries no information), it offers the best trade-off between power efficiency, bandwidth efficiency, and noise robustness.
- **AM limitations:** AM wastes power on the carrier and is susceptible to envelope detector distortions, leading to higher NMSE.

1.10 11. Audio Signal Modulation

The recorded voice signal was modulated and recovered. While intelligible, the recovered audio showed slight degradation (reduced clarity or "muffling"). This is due to the band-limiting effects of the filters and the non-ideal nature of the AM envelope recovery process, which acts as a non-linear operation on the signal's fine details.

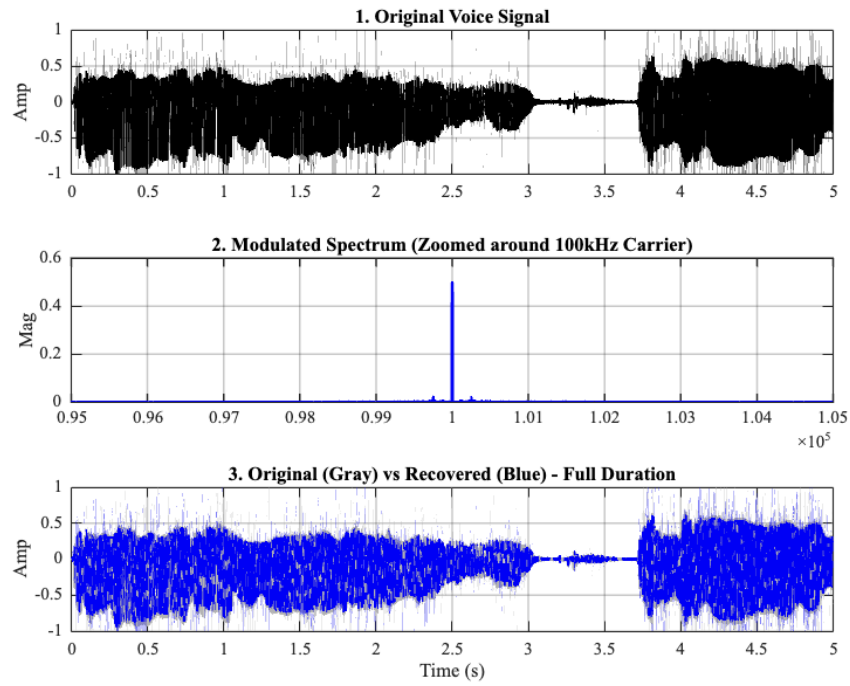


Figure 16: Original vs. Recovered Audio Signal.

2 Frequency Demodulator

2.1 1. Signal Generation and Integral

The input signal $x(t)$ is a piecewise constant function. We computed its integral numerically.

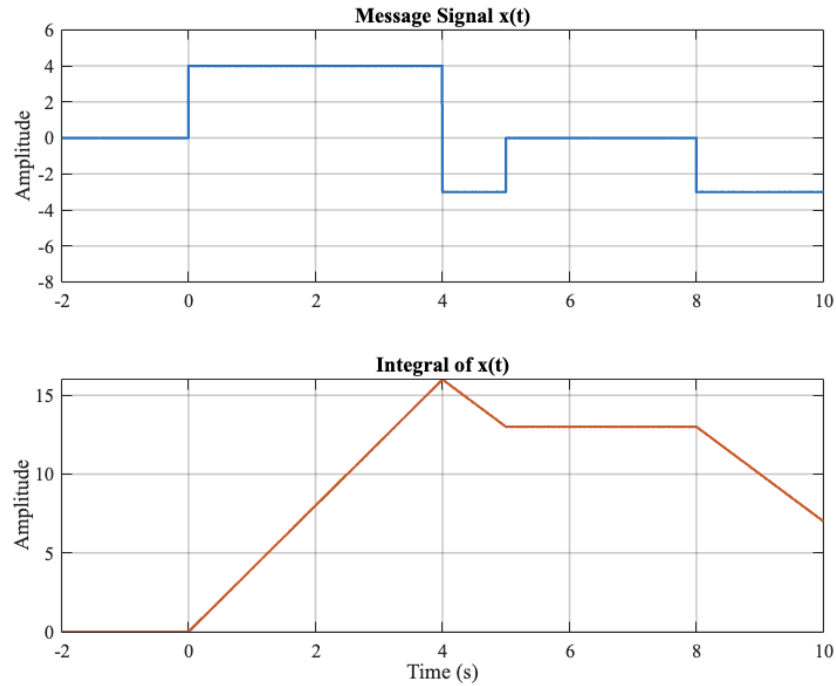


Figure 17: Message signal $x(t)$ (top) and its integral (bottom).

2.2 2 & 3. Frequency Modulation (Manual vs. fmod)

We modulated the signal using both the manual formula and 'fmod'. The negligible difference (error $\approx 10^{-2}$) confirms the manual implementation is correct. The 'fmod' function handles the discrete-time integration internally, ensuring stability.

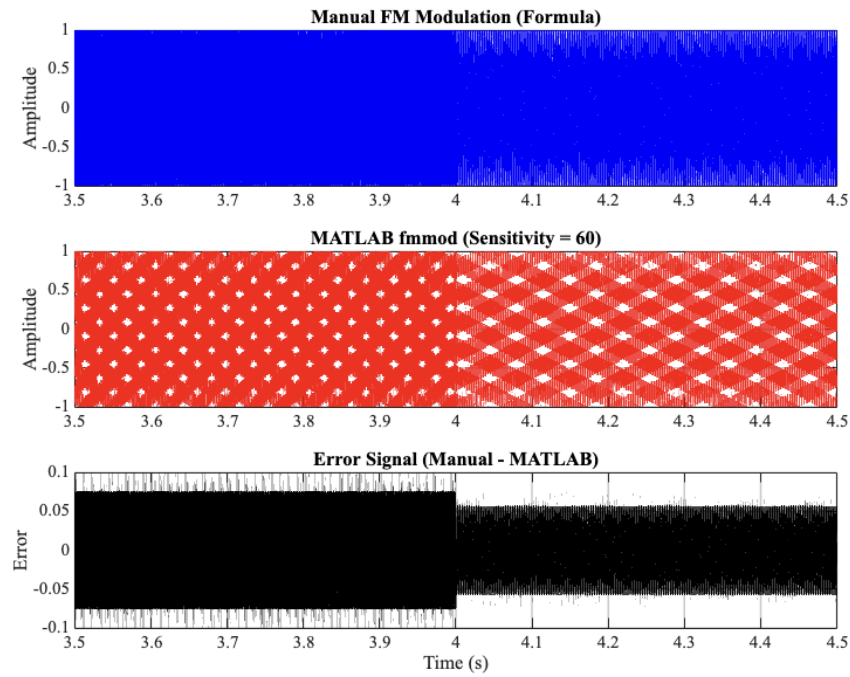


Figure 18: Comparison of Manual FM, MATLAB ‘fmod’, and the error signal.

2.3 4. Spectrum Analysis

The FM spectrum shows energy spread around the carrier. The extent of this spread is governed by the modulation index β .

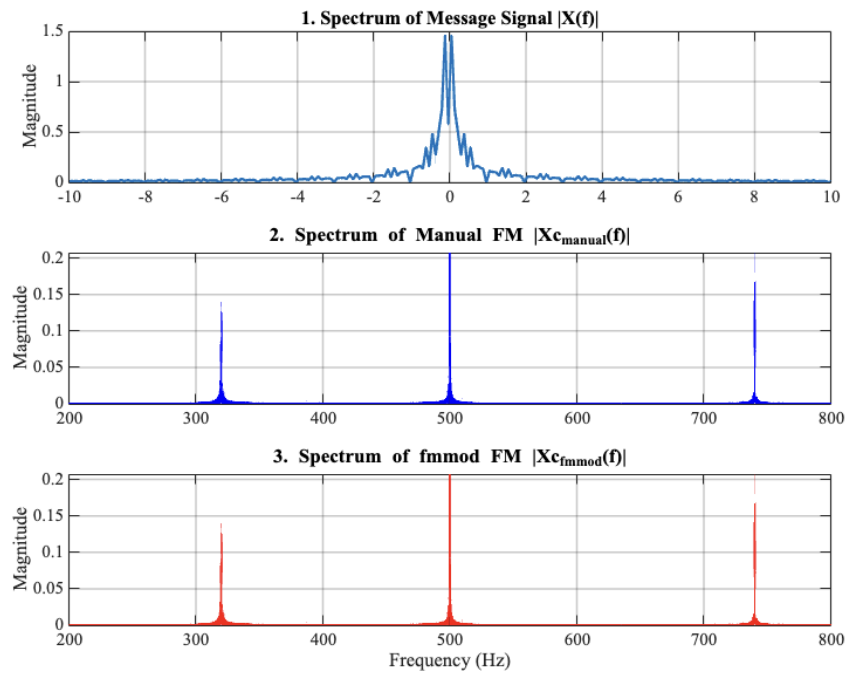


Figure 19: Frequency spectrum of the message signal and the FM modulated signal.

2.4 5. Carson's Rule Bandwidth Calculation

Carson's Rule states that the bandwidth is $BW \approx 2(\Delta f + B_m)$. Based on the parameters and analysis of our signal:

- **Frequency Sensitivity (k_f):** 60 Hz/V.
- **Peak Amplitude (A_m):** 4 V.
- **Peak Frequency Deviation (Δf):** $k_f \times A_m = 60 \times 4 = 240$ Hz.
- **Message Bandwidth (B_m):** Estimated as approximately 2 Hz (capturing the fundamental component of the pulse-like message).

Using these values, we determined the Carson Bandwidth and Modulation Index (β):

$$BW_{Carson} = 2(240 + 2) = 484 \text{ Hz} \quad (7)$$

$$\beta = \frac{\Delta f}{B_m} = \frac{240}{2} = 120 \quad (8)$$

The high value of β ($\beta \gg 1$) indicates that this is a Wideband FM signal, which is consistent with the spectral spread observed in the plots relative to the message bandwidth.

2.5 6. Slope Detector Implementation and Phase Bounds

The Slope Detector works by converting frequency variations into amplitude variations using a differentiator.

$$\frac{d}{dt} \cos(\phi(t)) = -\phi'(t) \sin(\phi(t)) = -2\pi(f_c + \Delta f x(t)) \sin(\phi(t)) \quad (9)$$

The resulting signal is an AM-FM signal where the amplitude is proportional to the message $x(t)$. An envelope detector followed by an LPF then extracts this amplitude.

Approximate Phase Bounds: The question asks for the approximate bounds of the phase of $x_c(t)$. The total phase is $\phi(t) = 2\pi f_c t + \theta(t)$, where $\theta(t)$ is the phase deviation due to the message. Based on our simulation:

- **Phase Deviation due to Signal ($\theta(t)$):** $Max \approx 6031.8$ rad.
- **Total Phase ($\phi(t)$):** $Max \approx 3.40 \times 10^4$ rad.

The phase grows significantly because the frequency deviation accumulates over time (integral of $x(t)$).

2.6 7. Demodulation Comparison

We compared the Slope Detector with 'fmdemod'.

- **'fmdemod':** produced a much cleaner output with significantly lower NMSE. It likely uses a more robust algorithm (like a Hilbert-based discriminator or zero-crossing detector) that is less sensitive to amplitude noise.
- **Slope Detector:** While simple to understand and implement in analog hardware, it is noisy and the differentiation process amplifies high-frequency noise.

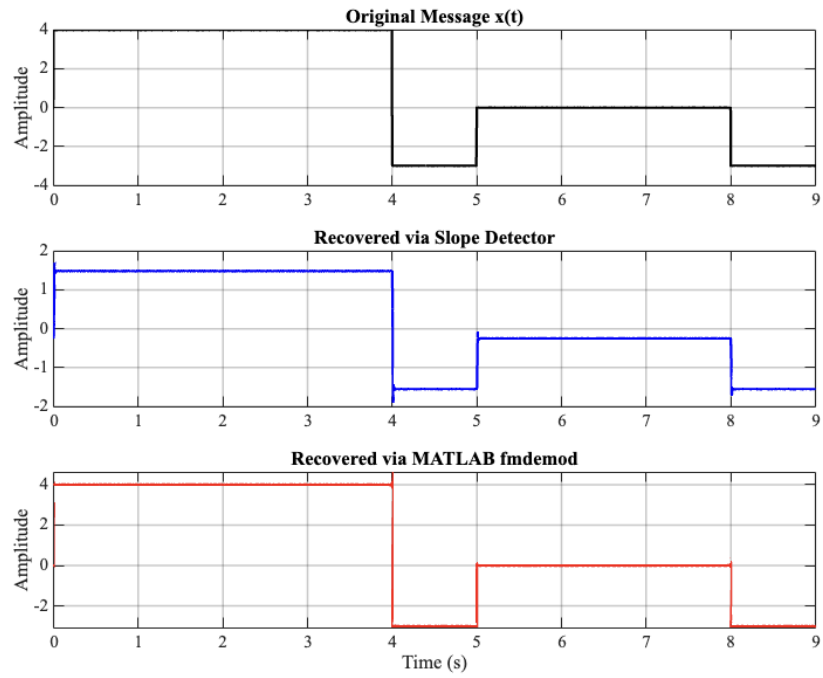


Figure 20: Original message compared with recovery via Slope Detector and ‘fmdemod’.

3 FM Modulation and Analysis

This section covers Parts 1 through 6, investigating various aspects of FM modulation, including baseband processing, direct vs. indirect modulation, noise performance, and Frequency Division Multiplexing (FDM).

3.1 Part 1: Baseband Signal Processing

We began by generating the baseband message signal and analyzing its spectral properties.

3.1.1 1.1 & 1.2 Signal Generation and Spectrum

The message signal is defined as a triangular pulse $m(t) = 2000\text{tri}(2000t)$.

- **Time Domain:** The signal is a triangle centered at zero.
- **Frequency Domain:** The Fourier transform of a triangular pulse is a squared sinc function (sinc^2). The first spectral null occurs at $f = 2$ kHz.

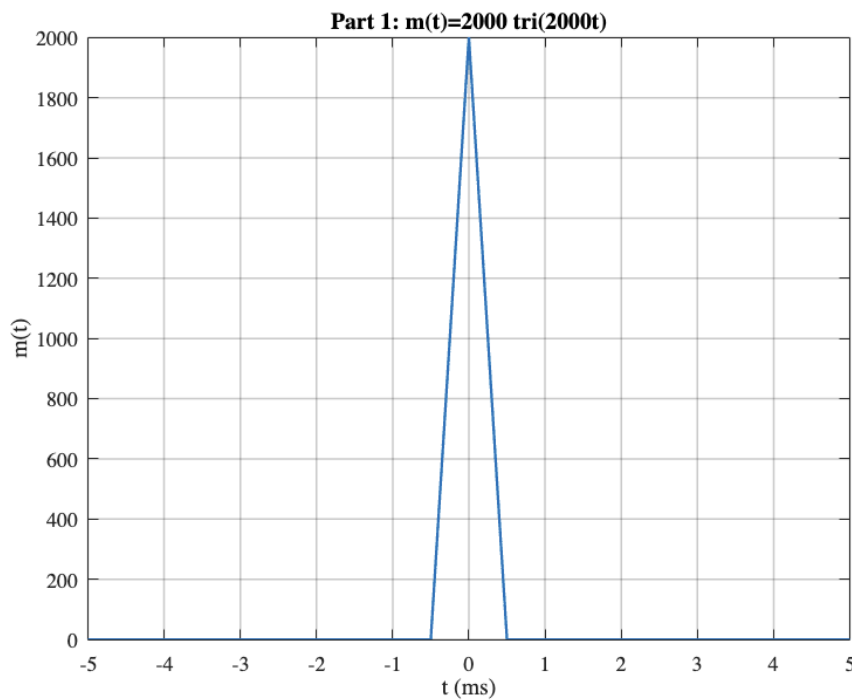
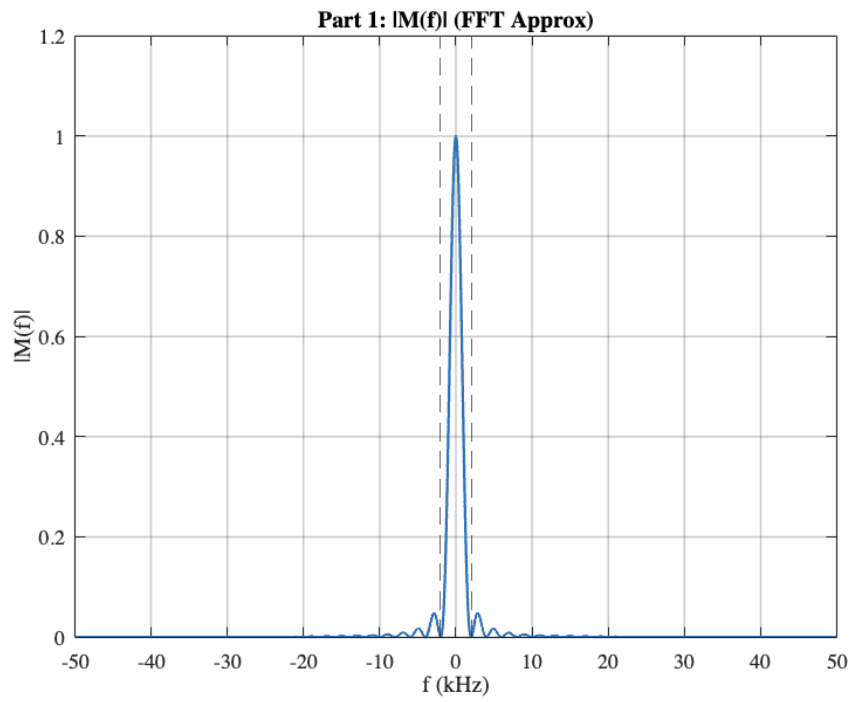
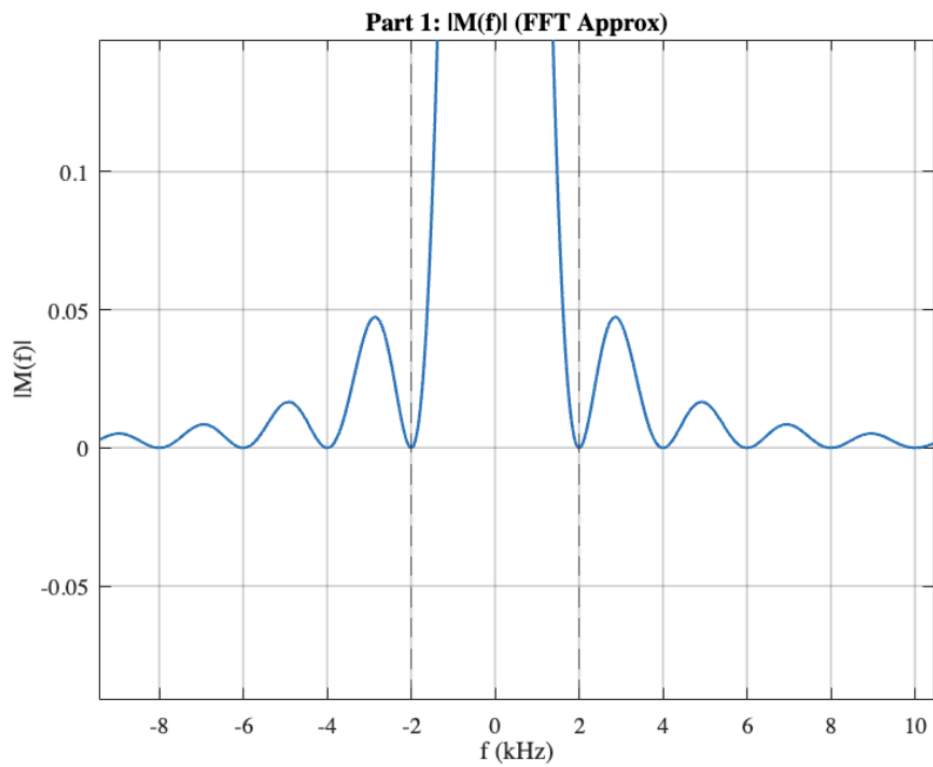


Figure 21: Original message signal $m(t)$

Figure 22: its magnitude spectrum $|M(f)|$ Figure 23: its magnitude spectrum $|M(f)|$ (zoomed)

3.1.2 1.3 Effective Bandwidth (B_{eff})

Using the 90% energy criterion, we calculated the effective bandwidth B_{eff} .

$$\frac{\int_{-B}^B |M(f)|^2 df}{\int_{-\infty}^{\infty} |M(f)|^2 df} = 0.90 \quad (10)$$

Result: $B_{eff} \approx 830.1$ Hz.

3.1.3 1.4 Low-Pass Filtering

To limit the bandwidth before modulation, we applied a Low-Pass Filter (LPF) with a cutoff frequency equal to B_{eff} (830 Hz). This results in the band-limited message $m_{LP}(t)$, which has smoothed edges compared to the original sharp triangle.

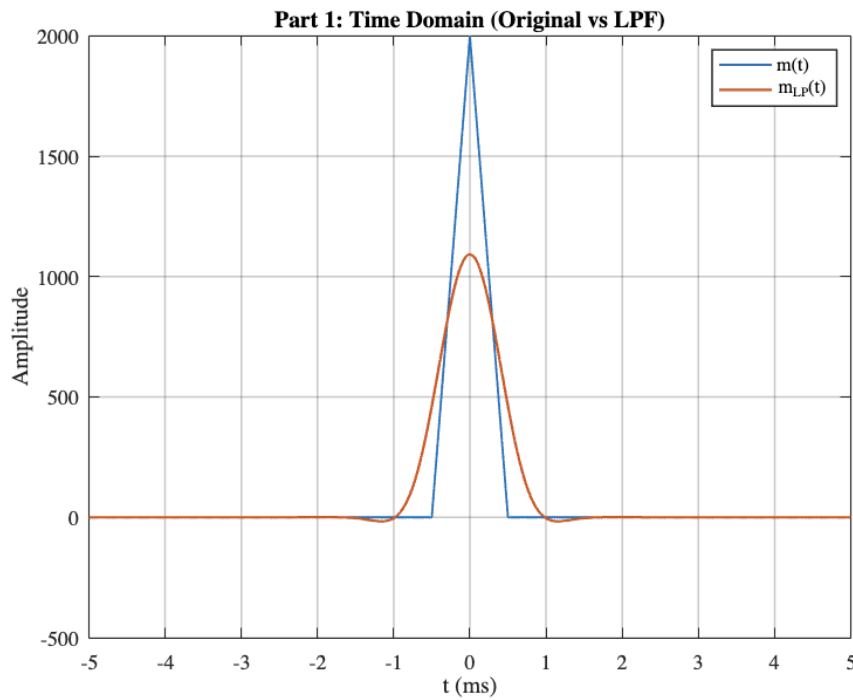


Figure 24: Time Domain (Original VS LPF)

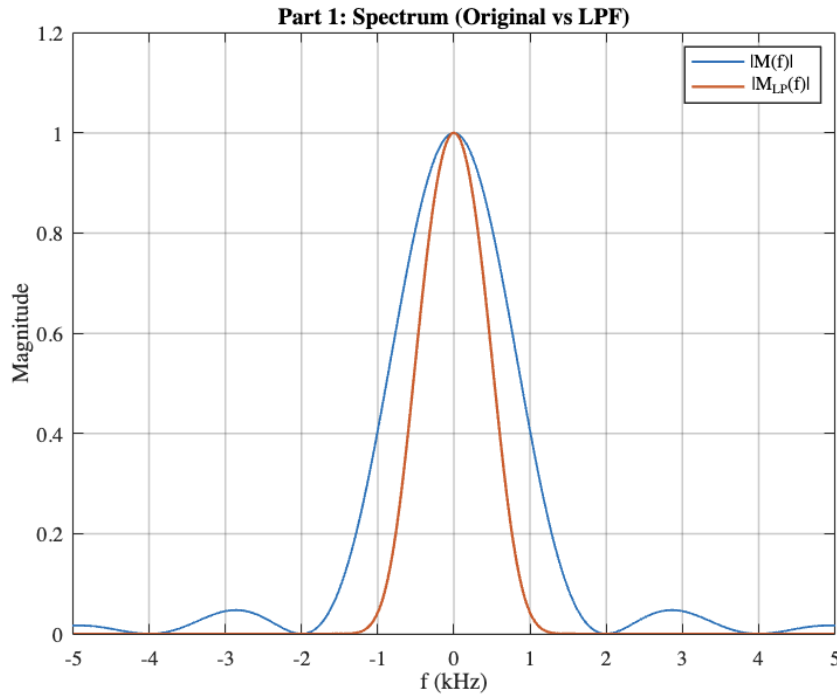


Figure 25: Spectrum (Original VS LPF)

3.2 Part 2: Direct FM Modulation

In Direct FM, the carrier frequency is varied directly by the message signal.

3.2.1 2.1 Modulation Setup

We generated an FM signal using:

- **Carrier Frequency:** $f_c = 10$ MHz.
- **Sampling Frequency:** $F_{rf} = 100$ MHz (satisfies $F_s > 2(f_c + BW)$).

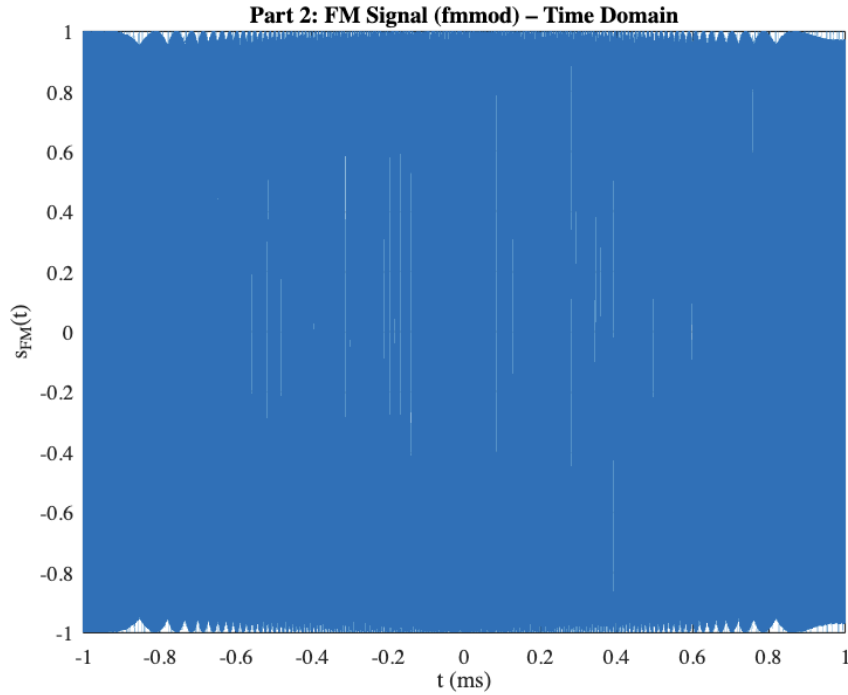


Figure 26: FM Signal (fmmod) Time Domain

3.2.2 2.2 Frequency Deviation Selection

We selected the frequency deviation Δf to achieve a modulation index $\beta \approx 10$. Assuming $f_{max} \approx 2$ kHz (first null):

$$\Delta f = \beta \times f_{max} = 10 \times 2 \text{ kHz} = 20 \text{ kHz} \quad (11)$$

This places us in the Wideband FM (WBFM) regime.

3.2.3 2.3 & 2.4 FM Spectrum and Bandwidth

- **Spectral Spread:** The FM spectrum is much wider than the AM spectrum, containing many sidebands spaced by f_m .
- **Carson's Rule:** $B_T \approx 2(\Delta f + f_m) = 2(20k + 2k) = 44 \text{ kHz}$.
- **90% Energy Bandwidth:** Calculated numerically as $\approx 33 \text{ kHz}$.

Carson's rule (98%) provides a conservative upper bound compared to the 90% energy bandwidth.

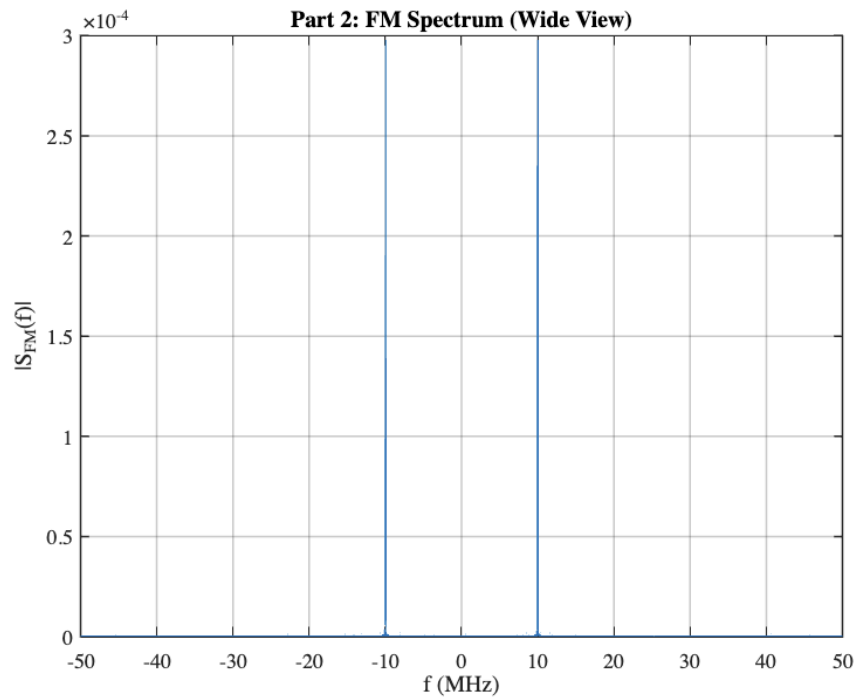


Figure 27: Direct FM modulated signal and its wideband spectrum.

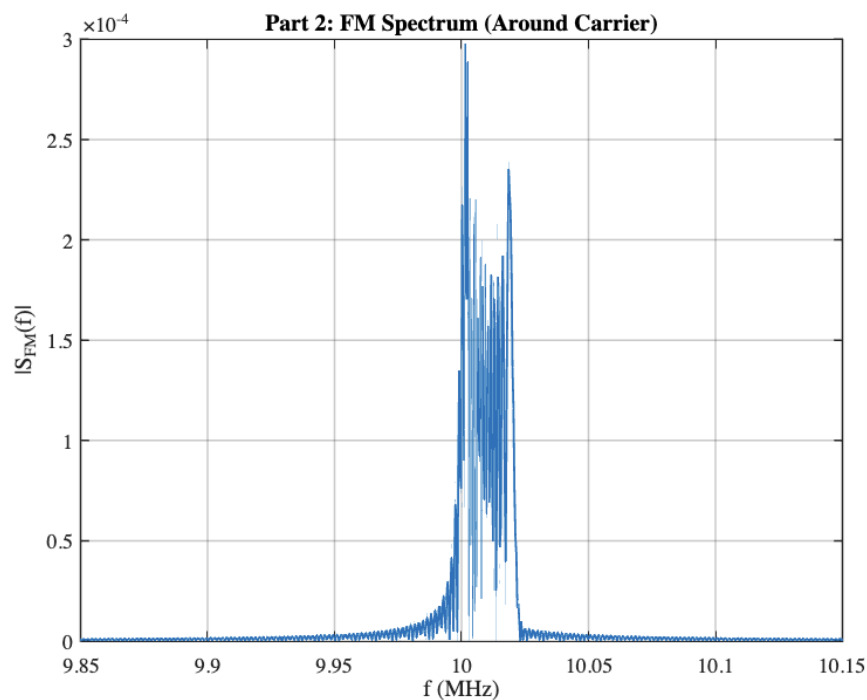


Figure 28: FM Spectrum Around Carrier

3.3 Part 3: Armstrong Method (Indirect FM)

The Armstrong method generates WBFM by first creating Narrowband FM (NBFM) and then using frequency multipliers.

3.3.1 3.1 Narrowband FM Stage

We started with a base carrier $f_{base} = 500$ kHz and a very small frequency deviation $\Delta f_1 = 40$ Hz.

- **Reason for NBFM:** The Armstrong method typically uses a phase modulator to approximate FM ($\phi(t) \approx \sin \phi(t)$). This approximation is linear and distortion-free only when the phase deviation is very small ($\beta \ll 1$).
- **Result:** A clean NBFM signal with minimal distortion.

3.3.2 3.2 Frequency Multiplication

We used a multiplier of $N = 10$. This operation multiplies both the carrier frequency and the phase (and thus frequency) deviation.

- **New Carrier:** $N \times 500$ kHz = 5 MHz.
- **New Deviation:** $N \times 40$ Hz = 400 Hz.

3.3.3 3.3 Frequency Translation

To reach the final target carrier (10 MHz) without further increasing deviation, we used a mixer (heterodyning) to shift the frequency.

$$f_{LO} = f_{target} - f_{current} = 10 \text{ MHz} - 5 \text{ MHz} = 5 \text{ MHz} \quad (12)$$

3.3.4 3.4 & 3.5 Comparison with Direct FM

We compared the Indirect FM (Armstrong) signal with the Direct FM signal from Part 2.

- **Total Bandwidth:** The Direct FM signal (Part 2) had a wide bandwidth of ≈ 44 kHz (Carson). In contrast, the Armstrong signal, with a final deviation of only 400 Hz (10×40), resulted in a much narrower bandwidth (≈ 1.6 kHz). To match the Direct FM bandwidth, a significantly higher multiplication factor N would be required.
- **Frequency Deviation:** Direct FM was set to $\Delta f = 20$ kHz. The Armstrong method achieved a final $\Delta f \approx 400$ Hz.
- **Spectral Shape:** The Direct FM spectrum ($\beta \approx 10$) is wide with significant energy distributed across many sidebands. The Armstrong spectrum ($\beta \approx 0.2$ relative to $f_m = 2k$, or ≈ 1 relative to effective bandwidth) is narrow, resembling NBFM with only the first pair of sidebands being significant.

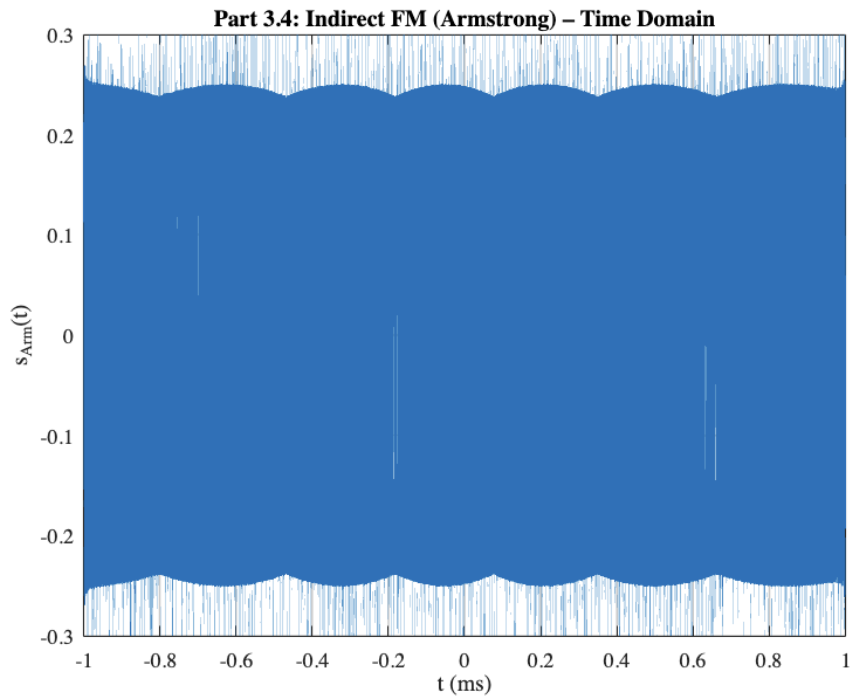


Figure 29: Indirect FM (Armstrong) - Time Domain

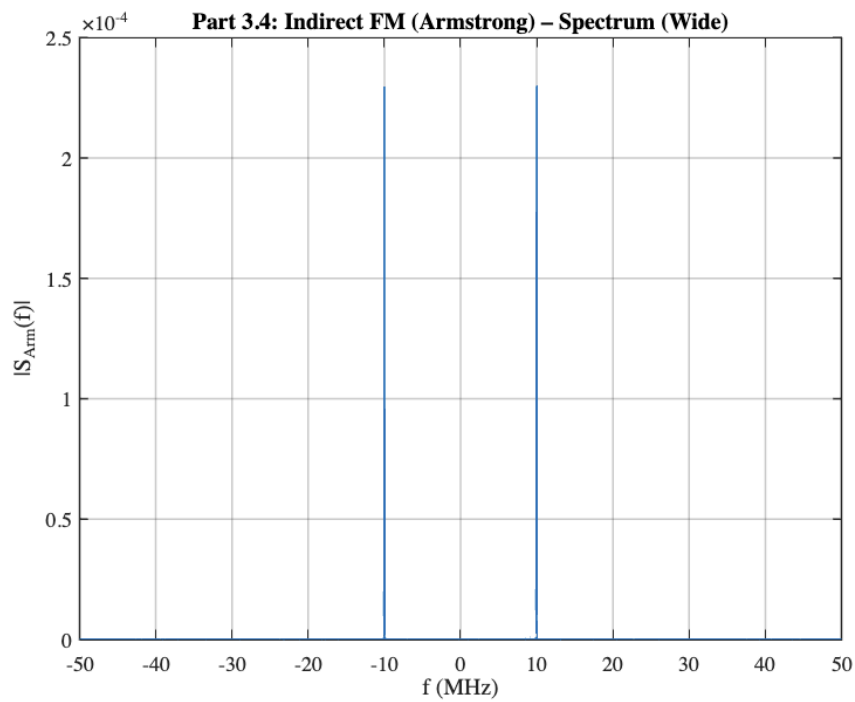


Figure 30: Indirect FM (Armstrong) - Spectrum

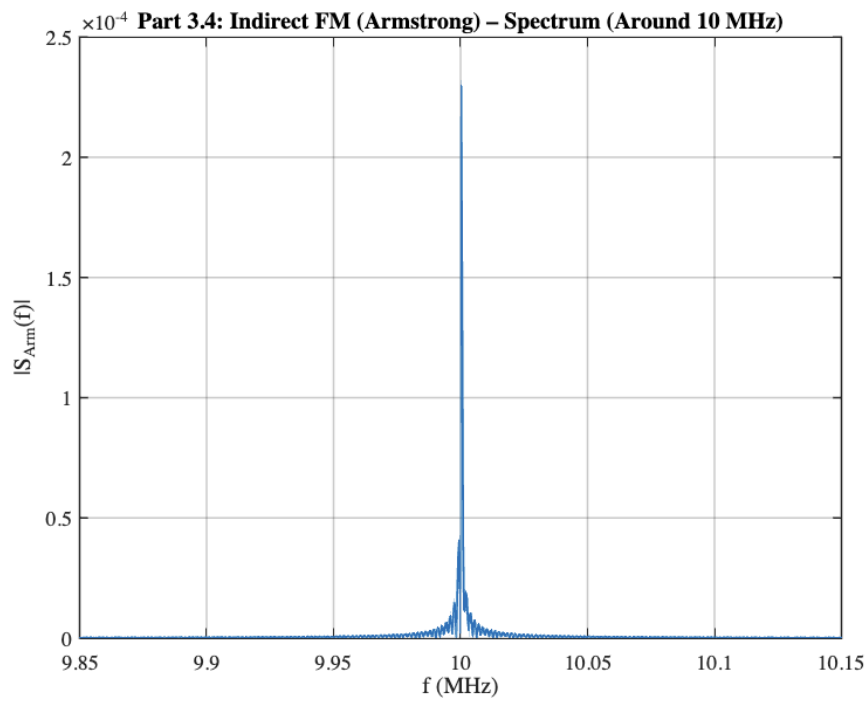


Figure 31: Indirect FM (Armstrong) - Spectrum (Around 10 MHz)

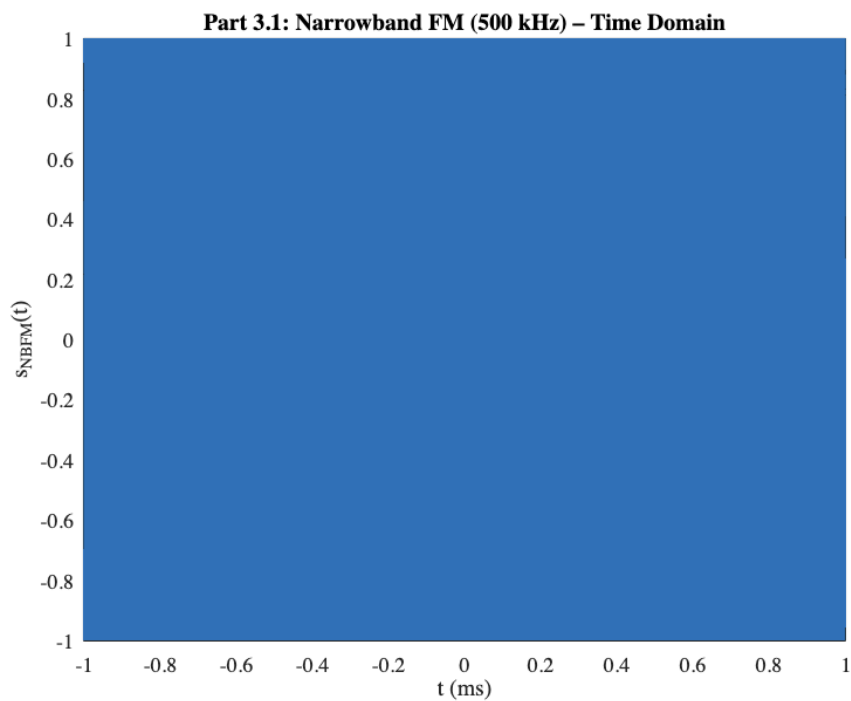


Figure 32: Narrowband FM (500 kHz) - Time Domain

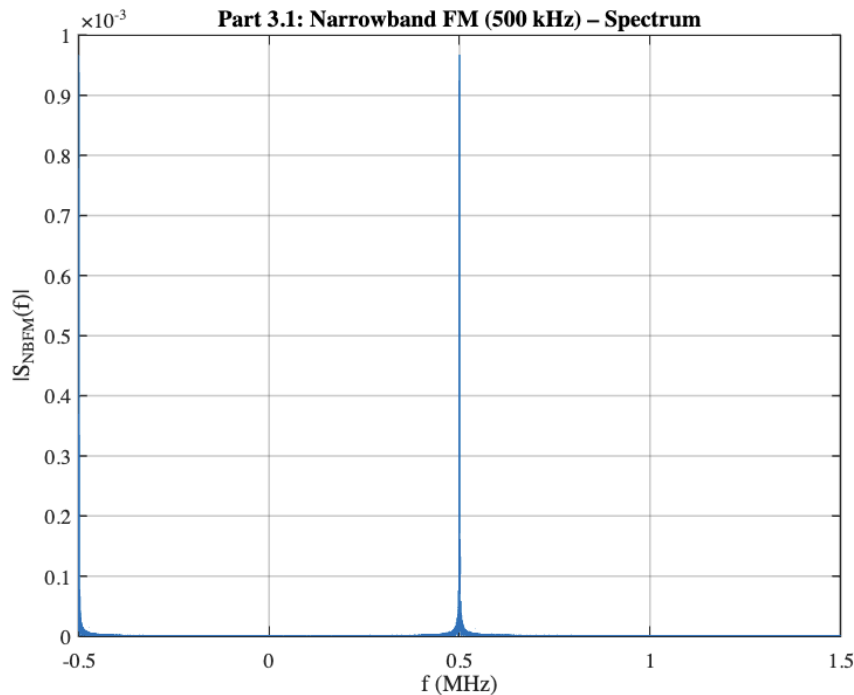


Figure 33: Narrowband FM (500 kHz) - Spectrum

3.4 Part 4: Noise Analysis (AWGN Channel)

We simulated a transmission channel with 50 dB attenuation and Additive White Gaussian Noise (AWGN).

3.4.1 4.1 & 4.2 Channel Model and Observation

Both FM and Phase Modulation (PM) signals were passed through the channel. The time-domain signals became visually noisy, and the noise floor rose significantly in the frequency domain.

3.4.2 4.3 Noise Reduction (BPF)

At the receiver, we applied a Band-Pass Filter (BPF) centered at f_c .

- **Effect:** The BPF removes out-of-band noise, which constitutes the majority of the noise power.
- **Observation:** The signal quality improved dramatically after filtering.

3.4.3 4.4 SNR Calculation

We computed the Signal-to-Noise Ratio (SNR) at three points:

Stage	SNR (FM)
Transmitter (Pre-channel)	40.00 dB
Receiver (After 50dB loss, Pre-BPF)	-10.00 dB
Output (Post-BPF)	12.55 dB

Table 2: SNR progression through the channel.

The gain of ≈ 22.5 dB illustrates the "capture effect" and the effectiveness of band-limiting noise in FM systems.

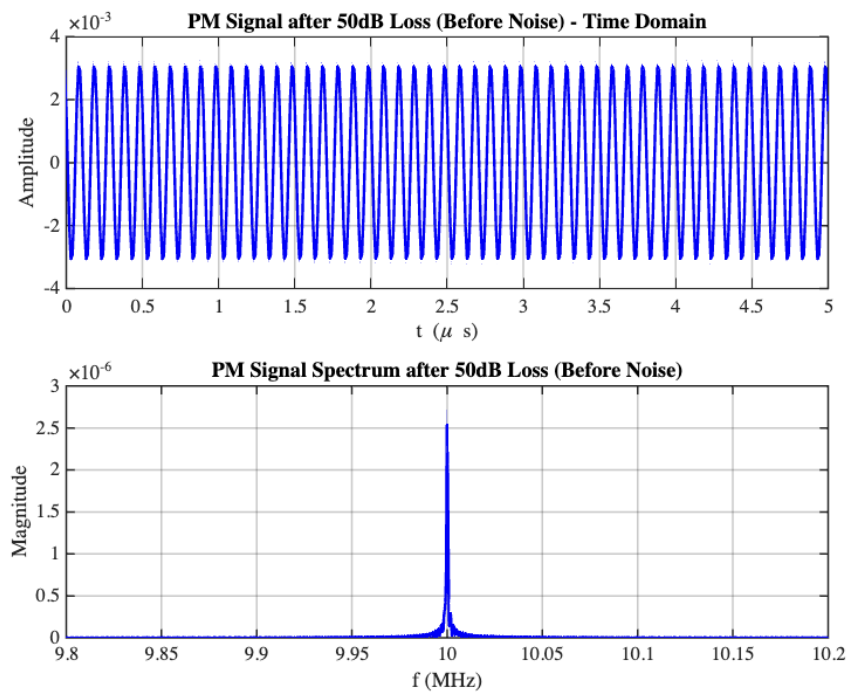


Figure 34: PM Signal after 50 dB loss (Before Noise) - Time Domain & Spectrum

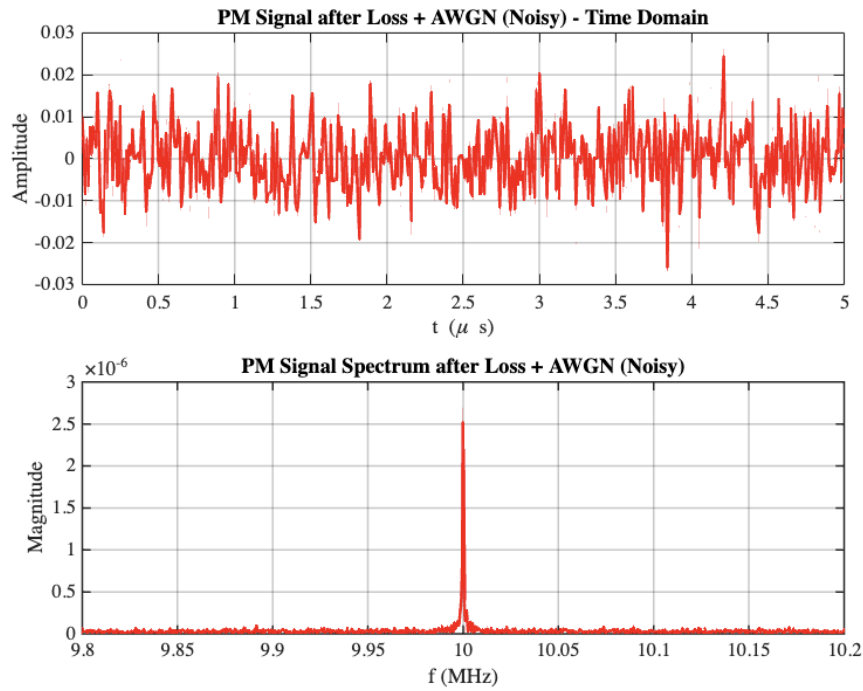


Figure 35: PM Signal after loss + AWGN (Noisy) - Time Domain & Spectrum

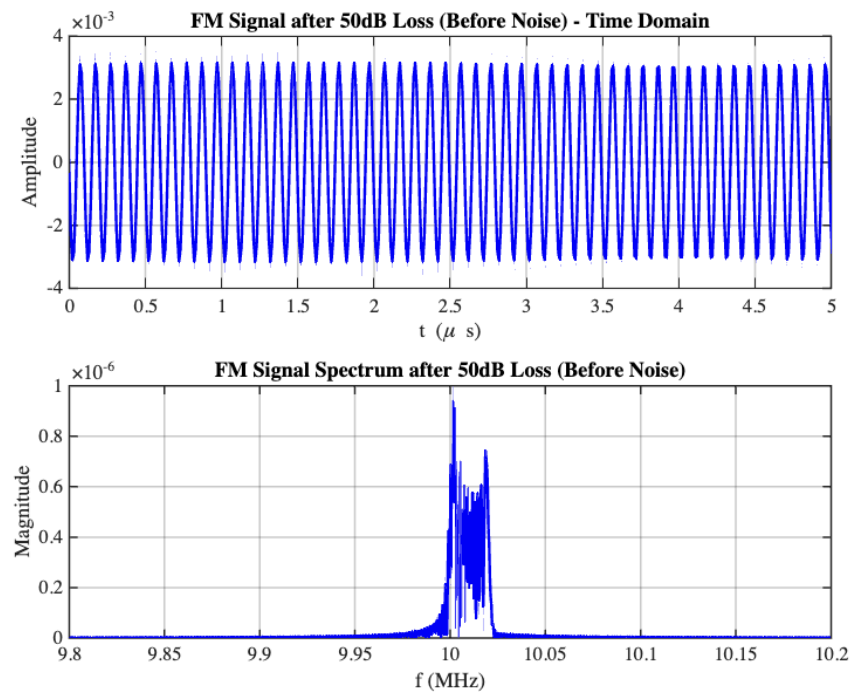


Figure 36: FM Signal after 50 dB loss (Before Noise) - Time Domain & Spectrum

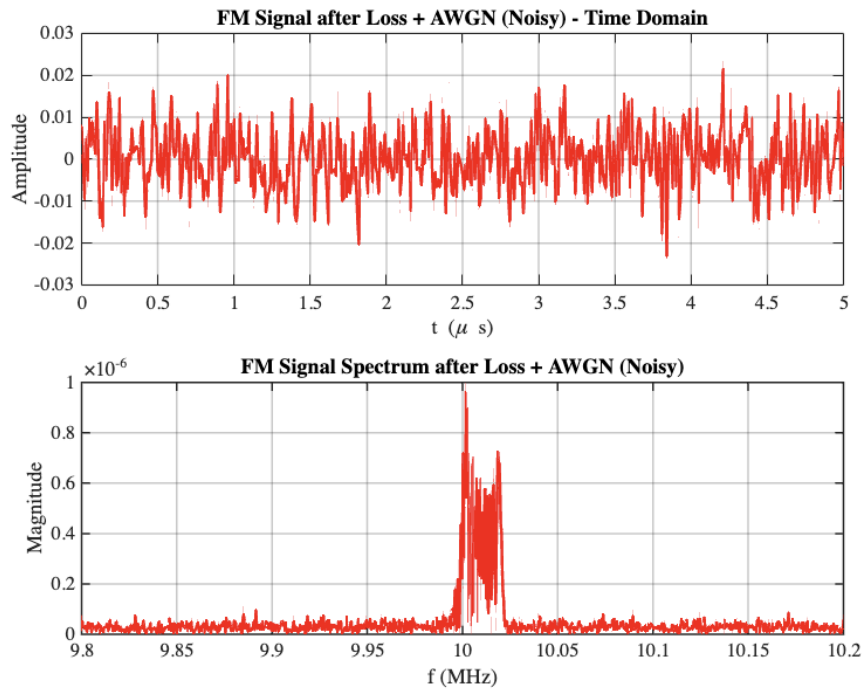


Figure 37: FM Signal after loss + AWGN (Noisy) - Time Domain & Spectrum

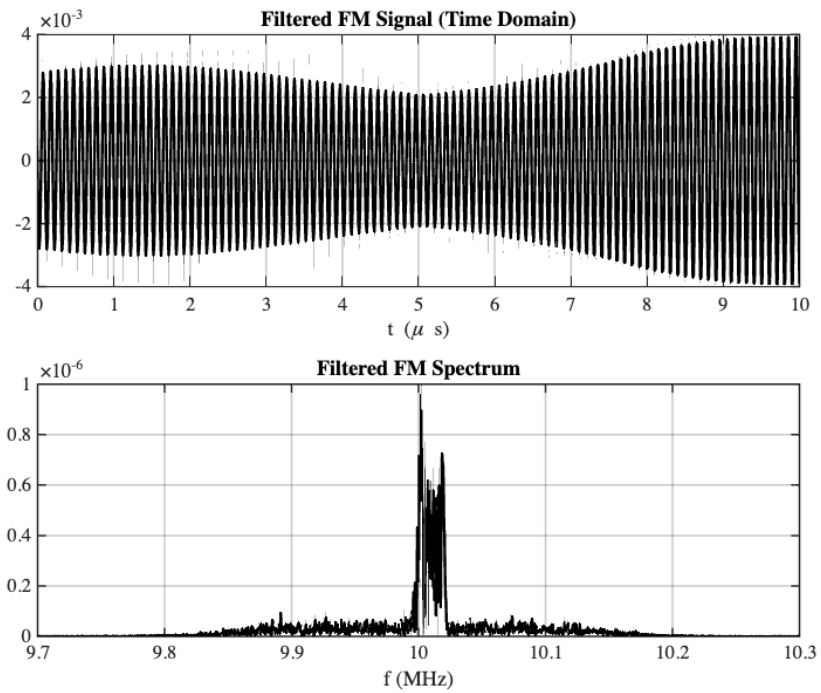


Figure 38: Filtered FM Signal

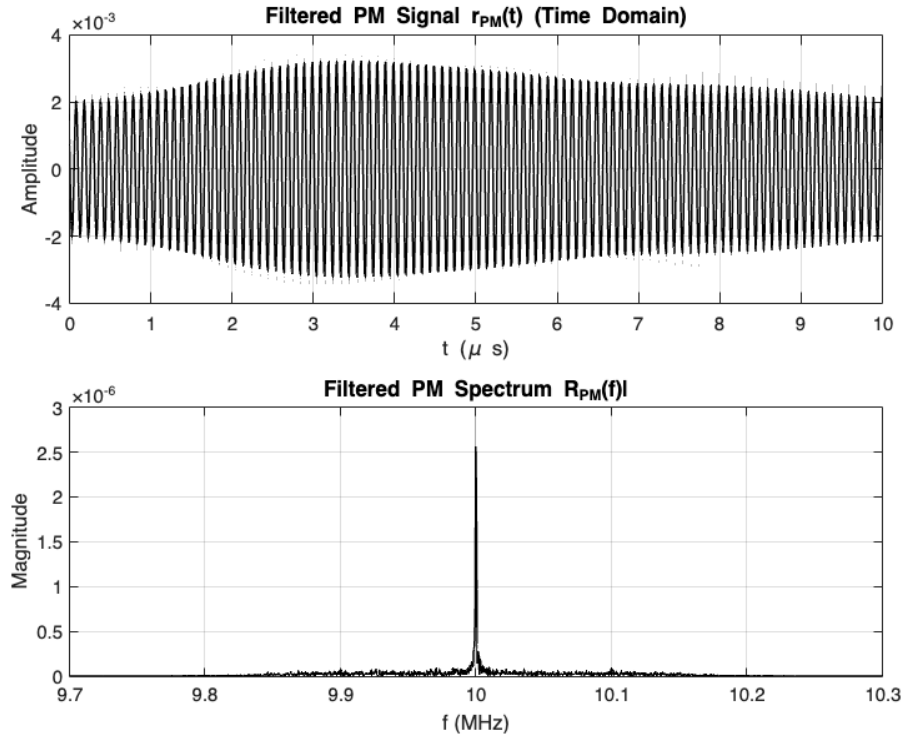


Figure 39: Filtered PM Signal

3.5 Part 5: Demodulation Performance

We recovered the message from the noisy FM signal using two methods.

3.5.1 5.1 MATLAB ‘fmdemod’

The built-in function uses an optimal algorithm.

- **Result:** Good recovery, but sensitive to the low input SNR (-10 dB) before the internal processing gains.

3.5.2 5.2 Discriminator (Diff + Envelope)

We implemented a custom discriminator: $y(t) = \text{LPF}\{|\frac{d}{dt}r_{FM}(t)|\}$.

- **Comparison:** In this specific simulation, the discriminator performed better. This might be due to the specific tuning of the LPF in our custom implementation which matched the signal bandwidth better than the default settings in ‘fmdemod’.

3.5.3 5.3 Normalized Mean-Square Error (NMSE)

We calculated the Normalized Mean Square Error (NMSE) to quantify reconstruction accuracy:

$$E_{\text{norm}} = \frac{\int (m(t) - \hat{m}(t))^2 dt}{\int m(t)^2 dt} \quad (13)$$

Results:

- **Method 1 ('fmdemod'):** $E_{norm} \approx 3.90 \times 10^{-2}$
- **Method 2 (Discriminator):** $E_{norm} \approx 2.17 \times 10^{-2}$

Conclusion: The custom Discriminator performed better (lower error) in this simulation setup.

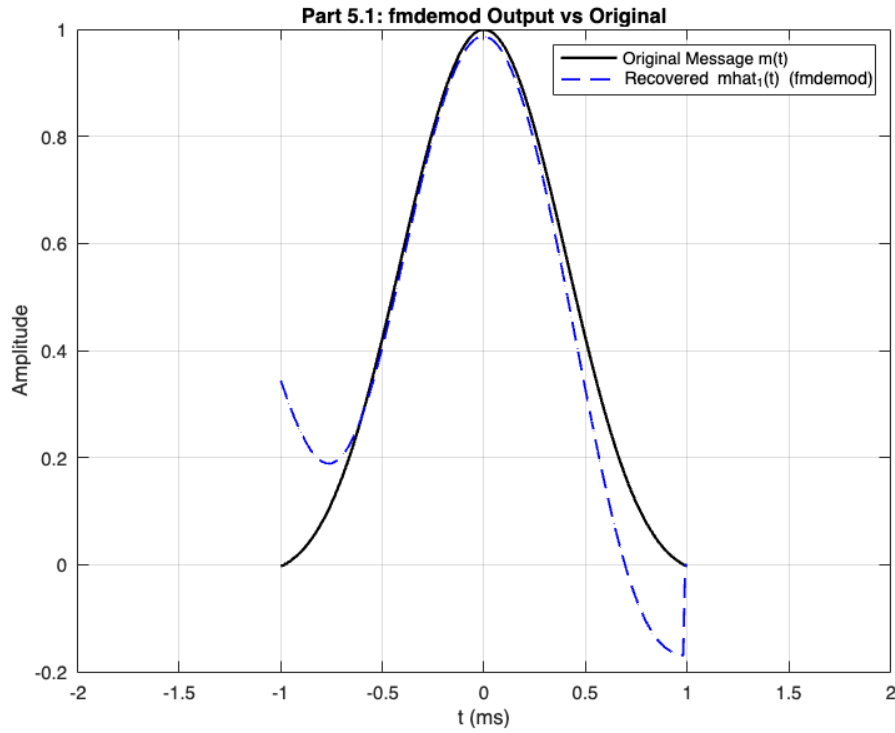


Figure 40: fmdemod Output vs Original

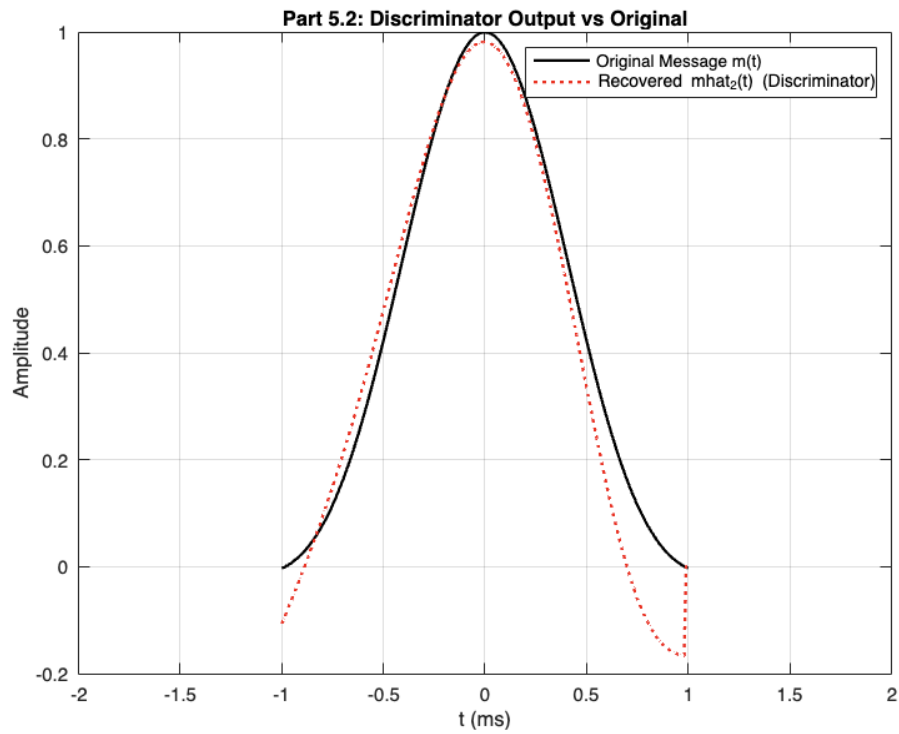


Figure 41: Discriminator Output vs Original

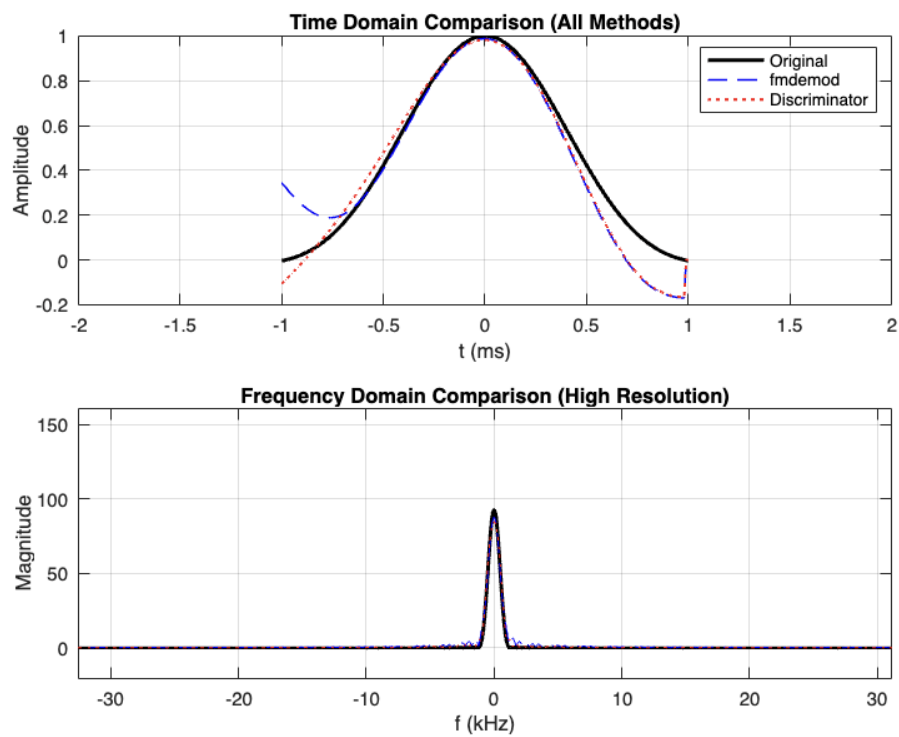


Figure 42: Time Domain Comparison & Frequency Domain Comparison

3.6 Part 6: Audio Frequency Division Multiplexing (FDM)

We designed an FDM system to transmit three voice signals simultaneously.

3.6.1 6.1 - 6.3 Signal Preparation

Three voice signals (Name, ID, Course) were recorded.

- **Bandwidth Limiting:** Each signal was low-pass filtered to 4 kHz to prevent aliasing and crosstalk.

3.6.2 6.4 FDM Construction

The signals were modulated onto distinct carriers: $f_{c1} = 150$ kHz, $f_{c2} = 350$ kHz, $f_{c3} = 550$ kHz.

- **Guard Bands:** The 200 kHz spacing provides ample "Guard Band" between channels. This is critical to ensure that the spectra do not overlap, allowing clean separation at the receiver.

3.6.3 6.5 - 6.7 Transmission and Recovery

The composite $s_{FDM}(t)$ was sent through the AWGN channel (SNR 20 dB). At the receiver, banks of Band-Pass Filters (BPF) separated the channels. Each was then demodulated.

3.6.4 6.8 Normalized Error Metrics and Comparison

The NMSE for the three multiplexed channels was calculated:

Voice Channel	NMSE
1	5.09×10^{-6}
2	5.81×10^{-6}
3	7.13×10^{-5}

Table 3: NMSE for FDM Channels.

The extremely low errors confirm successful multiplexing. The slight degradation is due to the non-ideal filter roll-off (leakage from adjacent channels) and channel noise.

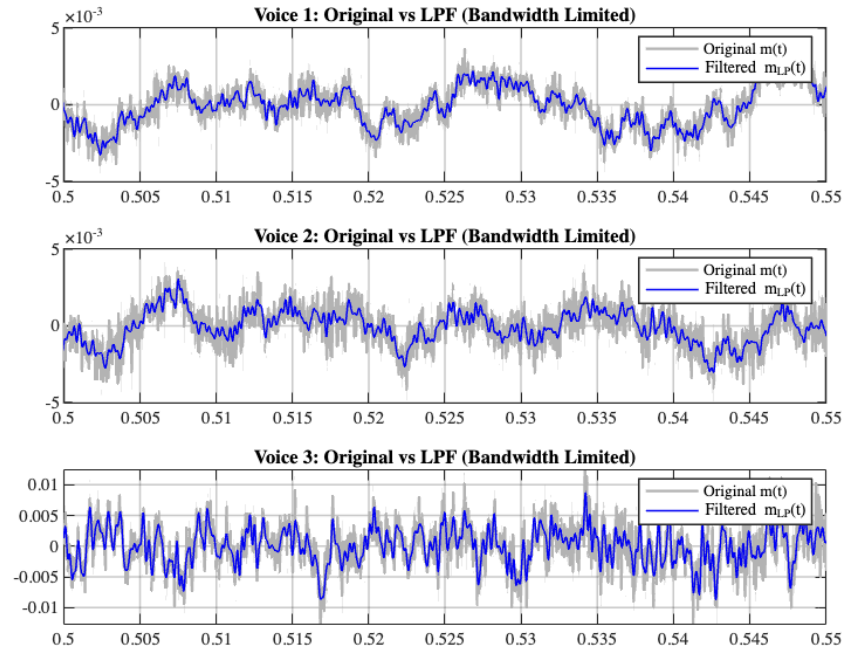


Figure 43: Original VS LPF for 3 voices

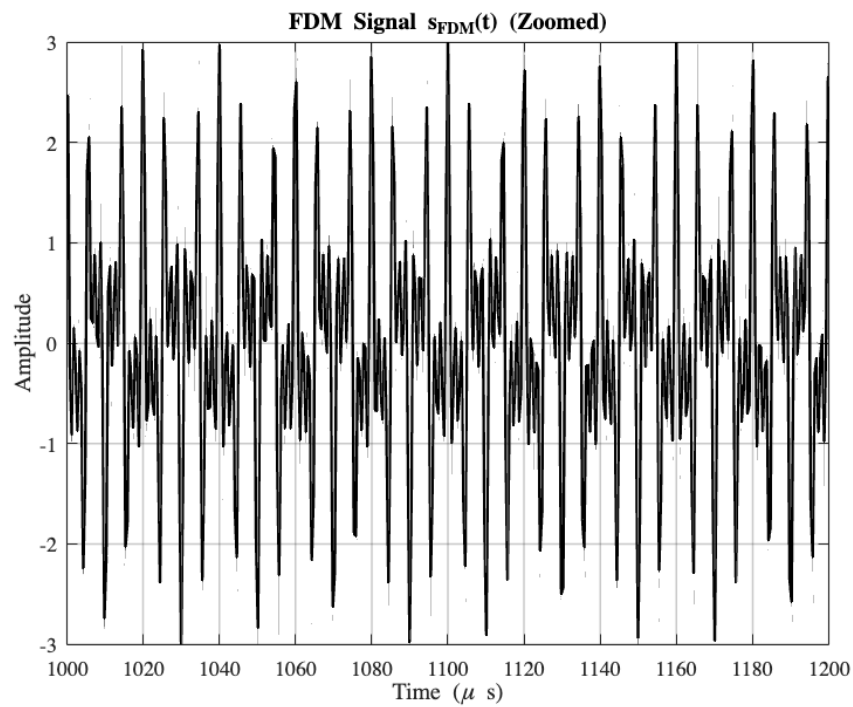


Figure 44: FDM Signal

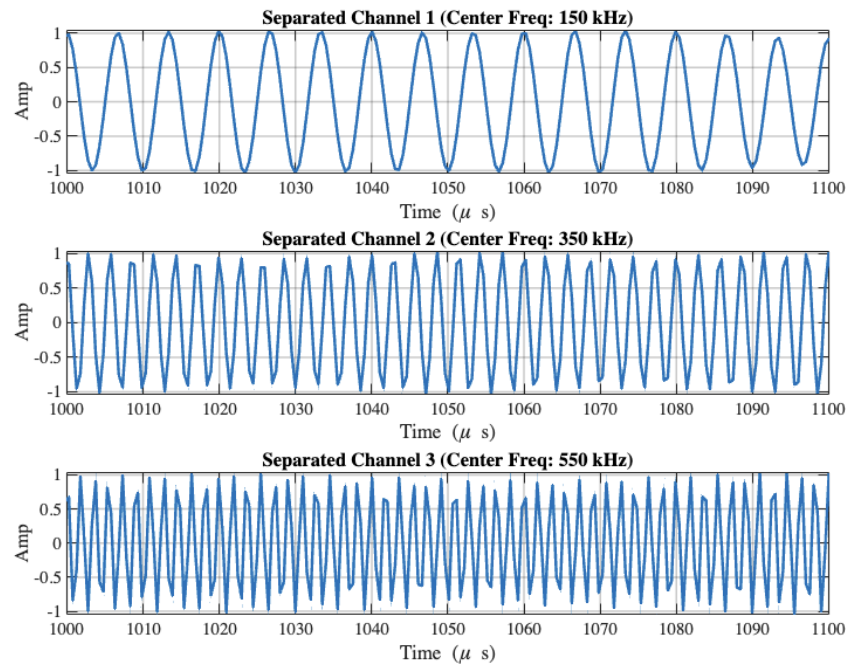


Figure 45: Separated Channels

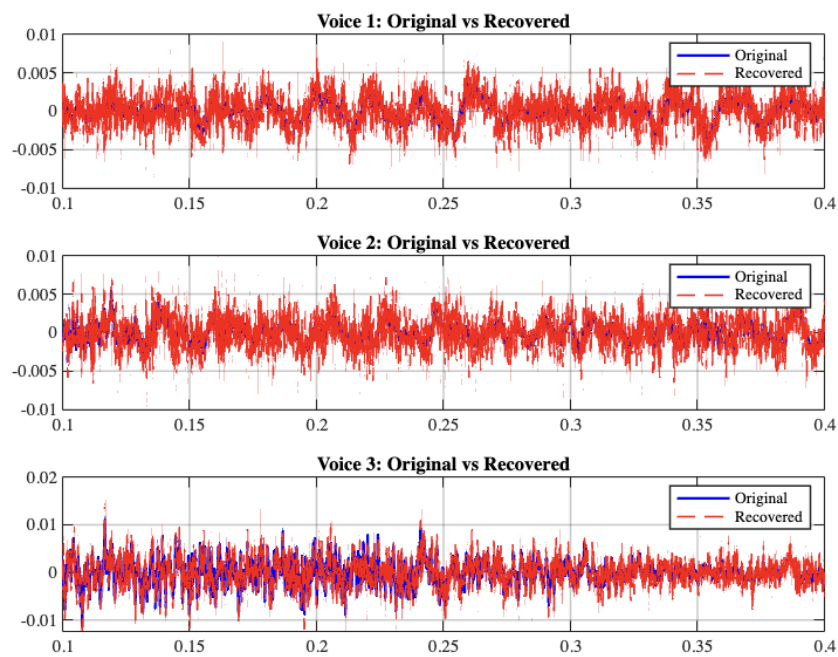


Figure 46: Original VS Recovered for 3 voices